Data Structures and Algorithms · 21CSC201J

Unit 3 · Session 6 · SLO 2 — Solved Worksheet

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Unit â€“ 3 : Session â€“ 6 : SLO â€“ 2 | Stack Applications â€” Tower of Hanoi

I. Fill in the Blanks

• TOWER OF HANOI is a classic problem that demonstrates the use of recursion with stacks.
• The minimum number of moves required to solve the Towers of Hanoi with n disks is 2^n - 1.
• The recursive solution to Towers of Hanoi uses the STACK data structure to manage function calls.
• In the Towers of Hanoi, disks are moved from the source rod to the DESTINATION rod using an auxiliary rod.
• Each recursive call pushes a new STACK FRAME (activation record) onto the stack.
• The Towers of Hanoi involves TWO recursive steps (move n-1 disks twice, plus one move).
• The stack unwinds when the BASE case of the recursive function is reached.
• The base condition in Towers of Hanoi is when ONE (n = 1) disk is to be moved.
• Stacks in recursion help to TRACK (preserve) the order of operations in solving the puzzle.
• The auxiliary rod in the Towers of Hanoi acts as a TEMPORARY STORAGE (buffer) during disk transfers.